

IRIN HOQUE ORCHI

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Research Interests

Natural Language Processing (NLP), Machine Learning, Large Language Models (LLMs), Low-Resource and Code-Mixed NLP, Uncertainty-Aware Learning, Human-Centered AI, Annotation Disagreement, Human-AI Interaction, Reliable and Trustworthy AI.

Currently investigating: whether displaying model confidence scores changes how human annotators anchor on suggested labels, with a particular focus on high-disagreement items.

Education

BRAC University

B.Sc. in Computer Science and Engineering

Dhaka, Bangladesh

December 2023

- GPA: 3.58/4.00 — Graduated with Distinction | IELTS: 7.0
- Relevant Coursework: Artificial Intelligence, Natural Language Processing, Data Structures & Algorithms, Database Systems, Software Engineering

Undergraduate Thesis

Mental Health Analysis of Cancer Patients with Lowest Survival Rates

2022–2023

Supervisor: Dr. Farig Yousuf Sadeque, Associate Professor, Dept. of CSE, BRAC University

- Constructed a novel dataset of 10,087 social media posts (Reddit, DailyStrength, HealthBoard) from cancer patients and caregivers across five cancer types selected for survival rates below 60%.
- Manually annotated 2,373 samples for emotional valence and arousal under multi-annotator agreement (Cohen's Kappa $\kappa = 0.40$ – 0.60 , moderate); introduced a meta-reviewer protocol to resolve labeling conflicts rather than discarding disagreement as noise.
- Designed and compared three NLP pipelines for 4-class emotion classification: **GRU (TF-IDF + FastText embeddings)** — **75% accuracy, highest overall**; few-shot SBERT (all-MiniLM-L6-v2) — 71% accuracy, F1 = 0.82, precision = 0.81, recall = 0.83, competitive under a severely constrained annotation budget; baseline SVM (Bigram + TF-IDF) — 62% accuracy.

Publications

[1] **Irin Hoque Orchi**. “A Qualitative Study of the Mental Health Experiences of Cancer Patients with the Lowest Survival Rate and Their Caregivers.” Proc. 26th IEEE International Conference on Computer and Information Technology (ICCIT 2023), Cox's Bazar, Bangladesh, December 2023. [**First Author**]
DOI: 10.1109/ICCIT60459.2023.10441282

[2] **Irin Hoque Orchi et al.** “Mitigating Online Harassment: Machine Learning Approaches for Hate Speech Detection in Transliterated Bengali Comments (Pandulipi).” Proc. 26th IEEE International Conference on Computer and Information Technology (ICCIT 2023), Cox's Bazar, Bangladesh, December 2023. [**Co-Author**]
DOI: 10.1109/ICCIT60459.2023.10441244

Manuscripts Under Review

[1] **I. H. Orchi**, T. Manzoor, T. A. Medha, A. Zaman, and S. R. Ramisa. “Class Imbalance in Regional Bangla Dialect Identification: A Benchmark Study with Interpretability Analysis.” Submitted to IEEE ICCIT 2026. [**First Author**]

[2] **I. H. Orchi**, T. Manzoor, A. Zaman, S. R. Ramisa, and F. Sadiq. “Adaptive Clarification: Evaluating LLM Uncertainty Signals for Ambiguity-Aware Dialogue.” Submitted to IEEE ICCIT 2026. [**First Author**]

[3] T. Manzoor, **I. H. Orchi**, and F. Sadiq. “AegisRL: An Adaptive Reinforcement Learning Framework for Alert Fatigue Mitigation in Security Operations Centers.” Submitted to IEEE ICCIT 2026. [**Co-Author**]

Research Experience

Class Imbalance in Regional Bangla Dialect Identification

2025

First Author

- Benchmarked class-weighting, oversampling, and SMOTE on BanglaDial (61,318 sentences, 12 classes) using LinearSVC and Logistic Regression; evaluated performance using Nadeau-Bengio variance-corrected resampled t-tests with Benjamini-Hochberg FDR control.
- Found standard imbalance handling produced no significant minority benefit while harming macro-F1 (0.756 baseline); audited cross-label duplication (2.33% corpus overlap) and evaluated a two-stage hierarchical classifier with oracle-routing controls, improving macro-F1 to 0.797.

- Formulated a 3-way response policy (ANSWER/HEDGE/CLARIFY) combining semantic entropy uncertainty estimation with a 17-feature query ambiguity classifier evaluated on AmbigNQ.
- Ran scaling experiments (60 vs. 1,200 questions, Qwen2.5-1.5B/7B-Instruct) using DeBERTa-v3 textual entailment clustering; found semantic entropy to be a weak, non-robust proxy for query ambiguity (AUC 0.596 → 0.517), while fusing response- and query-level features improved ambiguity classification (LOO macro-F1 0.450 → 0.571).

Pandulipi: Hate Speech Detection in Transliterated Bengali Comments

2022–2023

Co-Researcher

- Co-authored a study built on Pandulipi, the first publicly released transliterated Bengali hate speech corpus (5,000 Facebook comments; 2,837 hate / 2,163 non-hate), constructed by the project team, with 6 fine-grained categories.
- Contributed to comparing 7 classifiers across TF-IDF and Bag-of-Words representations; SVM with TF-IDF achieved the best binary result (accuracy = 0.743, F1 = 0.741), outperforming Bi-LSTM (accuracy = 0.576, F1 = 0.422).
- For multiclass classification, SVM achieved accuracy = 0.621 and F1 = 0.578, demonstrating the challenges of classification in short, noisy, code-mixed Bengali text.

Research Dataset

Mental Health Insights of Vulnerable Cancer Patients

Kaggle | Mendeley Data

- 10,087 raw text samples from Reddit, DailyStrength, and HealthBoard; 2,373 samples manually annotated for emotional valence and arousal.
- Inter-annotator reliability: Cohen’s Kappa $\kappa = 0.40\text{--}0.60$ (moderate); disagreements resolved through a meta-reviewer protocol.

Industry Experience

Software Engineer

August 2024 – Present

Bdjobs.com Limited

- Developing production web applications using ASP.NET Core, Angular, and RESTful APIs; contributing features from design through deployment.
- Building an AI-based candidate-job matching score feature that ranks candidate profiles against job requirements.

Intern, IT Division

November 2023 – February 2024

Dhaka Bank PLC

- Implemented biometric and face-recognition technology for employee attendance systems.

Technical Skills

ML & NLP Tools: PyTorch, Hugging Face Transformers, DeBERTa-v3, Qwen2.5, SBERT, FastText, GRU/LSTM, Scikit-learn, Imbalanced-learn (SMOTE)

NLP Tasks: Text Classification, Regional Dialect Identification, Hate-Speech/Emotion Detection, Hierarchical Classification, Cross-Label Corpus Auditing, TF-IDF (Word/Character N-grams), Corpus Annotation (Cohen’s Kappa)

Uncertainty & Dialogue: Semantic Entropy Quantification, Uncertainty Estimation, Ambiguity Classification, Selective Prediction / Response Policies (ANSWER/HEDGE/CLARIFY), Bidirectional Textual Entailment (NLI)

Statistical & Experimental Rigor: Nadeau-Bengio Corrected Resampled t-tests, Benjamini-Hochberg FDR Correction, Leave-One-Out CV (LOO-CV), Stratified Cross-Validation, ROC-AUC, Effect Size (Cohen’s d)

Languages & Tools: Python (primary), C++, TypeScript, Pandas, NumPy, Jupyter/Colab, Git, LaTeX, VS Code

Software Engineering: Angular, ASP.NET Core, RESTful APIs, SQL, Firebase

Presentations & Outreach

- Presented paper [1] as first author at IEEE ICCIT 2023, Cox’s Bazar, Bangladesh (December 2023).
- Delivered a talk on the Digital Equity Fellowship 2023 at Save the Children Bangladesh, addressing digital equity, ICT skills, and gender inclusion in technology.
- Presented live demonstrations of web applications and projects developed during professional work when they went into production at Bdjobs.com Limited.

Leadership & Activities

- Assistant Secretary, BRAC University Cultural Club (BUCUC) — 2020–2022
- Member, Photography Dept., BRAC University Adventure Club (BUAC) — 2020–2022
- Junior Executive, Finance & Marketing, Robotics Club of BRAC University (ROBU) — 2019–2022
- Volunteer, Kishor Alo, Prothom Alo — 2014–2018